Problem Set 2 - ECO6361 - Q01

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1. Goods Market Clearing Condition

Assumption: bond holdings $b_t = 0$ in equilibrium.

Implies that the HH does not hold bonds and therefore all savings go into capital or investment.

The household's BC:

$$c_t + inv_t + \frac{b_t}{1+r_t} + A(u_t)k_t = b_{t-1} + w_t n_t + r_t^k u_t k_t + \varpi_t$$

Since we assume $b_t = 0$ in EQ, this simplifies to:

$$c_t + inv_t + A(u_t)k_t = w_t n_t + r_t^k u_t k_t + \varpi_t$$

The firm's production function :

$$y_t = a_t k_t^\alpha n_t^{1-\alpha}$$

The firm maximizes profits:

$$\varpi_t = y_t - w_t n_t - r_t^k k_t$$

In equilibrium, **aggregate demand** (total consumption and investment) must equal **aggregate supply** (output from production). Accordingly, the goods market equilibrium condition is:

$$c_t + inv_t = y_t$$

Substituting the production function for y_t:

$$c_t + inv_t = a_t k_t^\alpha n_t^{1-\alpha}$$

The goods market equilibrium condition is:

$$c_t + inv_t = a_t k_t^\alpha n_t^{1-\alpha}$$